Assignment 1

1. Python program to check leap year

```
year = int(input("Enter year : "))
if (year % 4 == 0) or (year % 100 != 0) or (year % 400 == 0):
  print(year, " is a Leap Year!")
else:
  print(year, " is not a Leap Year!")
```

```
Enter year : 6000
6000  is a Leap Year!
```

2. Python Program to Find the Largest Among Three Numbers

```
num1 = int(input("Enter 1st number : "))
num2 = int(input("Enter 2nd number : "))
num3 = int(input("Enter 3rd number : "))
if num1 >= num2 and num1 >= num3:
  print(num1, " is largest number!")
elif num2 >= num1 and num2 >= num3:
  print(num2, " is largest numeber!")
elif num3 >= num1 and num3 >= num2:
  print(num3, " is largest number!")
```

```
Enter 1st number : 30
Enter 2nd number : 0
Enter 3rd number : 30
30  is largest number!
```

3. Python Program to Check if a Number is Positive, Negative or 0

```
num = int(input("Enter an Integer : "))
if num < 0:
  print(num, " is Negative!")
else:
  print(num, " is Positive!")
```

```
Enter an Integer : -30
-30  is Negative!
```

4. A toy vendor supplies three types of toys: Battery Based Toys, Key-based Toys, and Electrical Charging Based Toys. The vendor gives a discount of 10% on orders for battery-based toys if the order is for more than Rs. 1000. On orders of more than Rs. 100 for key-based toys, a discount of 5% is given, and a discount of 10% is given on orders for electrical charging based toys of value more than Rs. 500. Assume that the numeric codes 1,2 and 3 are used for battery based toys, key-based toys, and electrical charging based toys respectively. Write a program that reads the product code and the order amount and prints out the net amount that the customer is required to pay after the discount.

```
print("Product Codes")
print("1. Battery Based Toys")
print("2. Key Based Toys")
print("3. Electrical Based Toys")
code = int(input("Enter Product Code : "))
order = float(input("Enter Order Amount : "))

if code == 1 and order > 1000:
  net = order - (order * 0.10)
elif code == 2 and order > 100:
  net = order - (order * 0.05)
elif code == 3 and order > 500:
  net = order - (order * 0.10)
else:
  net = order

print("Net amount to be paid : ", net, "Rs.")
```

```
Product Codes
1. Battery Based Toys
2. Key Based Toys
3. Electrical Based Toys
Enter Product Code : 2
Enter Order Amount : 1000
```

```
Net amount to be paid :  950.0 Rs.
```

5. A transport company charges the fare according to following table:

Distance	Charges
1-50	8 Rs./Km
51-100	10 Rs./Km
> 100	12 Rs/Km

```
distance = float(input("Enter Distance in Km : "))
if distance <= 50 and distance >= 1:
  charges = distance * 8
elif distance >= 51 and distance <= 100:
  charges = distance * 10
elif distance > 100:
  charges = distance * 12
print("Transport Charges : ", charges, "Rs.")
```

```
Enter Distance in Km : 101.50
Transport Charges :  1218.0 Rs.
```

Assignment 2

1. Write a python program to reverse a number using a while loop.

```
num = int(input("Enter a number : "))
rev = 0
while (num > 0):
  rev = rev * 10 + (num % 10)
  num = num // 10

print("Reversed number : ", rev)
```

```
Enter a number : 1234
Reversed number :  4321
```

2. Write a python program to check whether a number is palindrome or not?

```
num = int(input("Enter a number : "))
og = num
rev = 0
while (num > 0):
  rev = rev * 10 + (num % 10)
  num = num // 10

if og == rev:
  print("Number is Palindrome!")
else:
  print("Number is not Palindrome!")
```

```
Enter a number : 12121
Number is Palindrome!
```

3. Write a python program finding the factorial of a given number using a while loop.

```
num = int(input("Enter a number : "))
fact = 1
while (num>0):
```

```
  fact *= num
  num -= 1
print("Factorial : ", fact)
```

```
Enter a number : 5
Factorial :  120
```

4. Accept numbers using input() function until the user enters 0. If user input 0 then break the while loop and display the sum of all the numbers.

```
sum = 0
while True:
  num = int(input("Enter number : "))
  if (num == 0):
    break
  sum += num
print("Sum of all numbers : ", sum)
```

```
Enter number : 10
Enter number : 9
Enter number : 8
Enter number : 7
Enter number : 6
Enter number : 5
Enter number : 4
Enter number : 3
Enter number : 2
Enter number : 1
Enter number : 0
Sum of all numbers :  55
```

Assignment 3

1. Print the first 10 natural numbers using for loop

```
for i in range(1,11):
  print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

2. Python program to check if the given string is a palindrome

```
s = input("Enter a string : ").lower()
og = s
rev = ''
for char in s:
  rev = char + rev

if og == rev:
  print("String is Palindrome!")
else:
  print("String is not Palindrome!")
```

```
Enter a string : Level
String is Palindrome!
```

3. Python program to check if a given number is an Armstrong number

```
num = int(input("Enter a number : "))
og = num
arm = 0
count = 0
while (num > 0):
  num = num // 10
  count +=1
num = og
```

```
while (num > 0):
  power = 1
  digit = (num % 10)
  num = num // 10
  for i in range(1, count+1):
    power = power * digit
  arm += power

if og == arm:
  print(og, " is an Armstrong number!")
else:
  print(og, " is not an Armstrong number!")
```

```
Enter a number : 153
153  is an Armstrong number!
```

4. Python program to get the Fibonacci series between 0 to 50

```
num1 = 0
num2 = 1
num3 = 0
print("Fibonacci series :")
print(num1)
print(num2)
while True:
  num3 = num1 + num2
  num1 = num2
  num2 = num3
  if num3 > 50:
    break
  print(num3)
```

```
Fibonacci series :
0
1
1
2
3
5
8
13
21
34
```

5. Python program to check the validity of password input by users

```
def validity_pass(password):
    uppercase = False
    lowercase = False
    digit = False
    special = False
    special_char = "!@#$%^&*()-_+=."

    if len(password) < 8:
        print("Password must be at least 8 characters long.")
        return

    for char in password:
        if char.isupper():
            uppercase = True
        elif char.islower():
            lowercase = True
        elif char.isdigit():
            digit = True
        elif char in special_char:
            special = True

    if not uppercase:
        print("Password must contain at least one uppercase letter.")
        return
    if not lowercase:
        print("Password must contain at least one lowercase letter.")
        return
    if not digit:
        print("Password must contain at least one digit.")
        return
    if not special:
        print("Password must contain at least one special character {!@#$%^&*()-_+=.}")
```

```
            return

        print("Password is valid ")


password = input("Enter Password: ")
validity_pass(password)
```

```
Enter Password: rofk
Password must be at least 8 characters long.
```